

APPENDIX A

Benchmark Specifications and Reproducibility

This appendix documents the benchmark harness configuration used in Chapters 3 and 4. All scenario prompts, metric definitions, and runner commands are reproduced so that independent researchers can replicate the reported outcomes on frozen notebook snapshots.

A.1 Harness Architecture

Benchmarks execute through three Python runners under `testing/host/scripts/`:

- `run_ask_tests.py` — Ask mode explanation suite;
- `run_code_tests.py` — Code mode generation suite;
- `run_agent_tests.py` — Agentic mode dispatch suite.

Each runner loads a JSON configuration file (`fyp_experiment_benchmarks_*.json`) and replays prompts through `streaming.py` with the `-live-llm` flag. Structured results are written to `testing/host/data/logs/`. Browser tools are mocked at the extension boundary during harness execution; trace logs record dispatched tool types and argument payloads without requiring a live Kaggle session.

The aggregate dissertation report is generated by:

```
python testing/host/scripts/generate_fyp_dissertation_report.py
```